Learning in the Age of AI

Lecture 02: Human Learning Psychology

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Memory Is Not Exposure

Guiding Question

Why does rereading often feel productive while retrieval practice produces stronger long-term learning?

$$\text{retrieval} > \text{re-exposure}$$

Speaker notes

Explain the distinction between familiarity and retrievability — students often confuse the two.

Why Might This Be True?

- rereading creates fluency
- fluency feels like understanding
- retrieval forces reconstruction

Interpretation

If you cannot bring the idea back without support, the idea is not stable yet ([?BrownRoedigerMcDaniel2014]; [?Dunlosky2013]).

Test-enhanced learning shows retrieval improves delayed retention, even when restudying looks better immediately after ([?RoedigerKarpicke2006]; [?KarpickeBlunt2011]).

Evidence

- **Immediate impression:** rereading creates familiarity
- **Delayed outcome:** retrieval produces stronger later recall
- **Didactic implication:** ask students to explain from memory before showing the polished answer

> The right test of learning is not recognition now, but reconstruction later.

Forgetting and Spacing

Guiding Question

Why can forgetting actually help learning?

$$R = e^{-t/s}$$

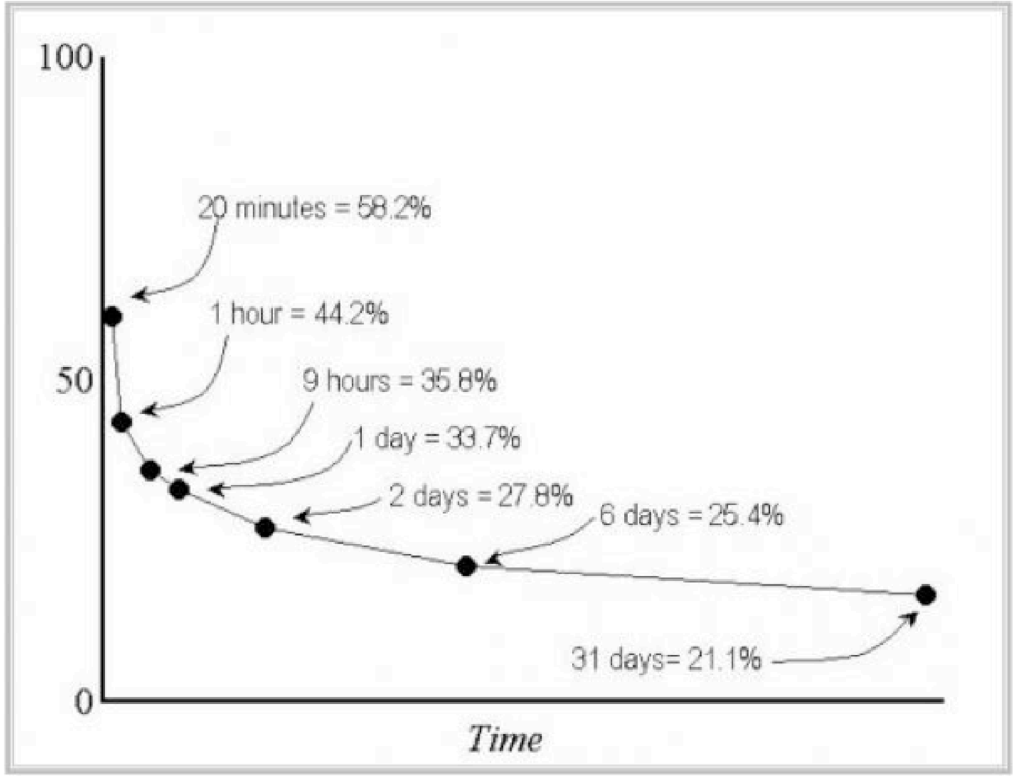
Why Might This Be True?

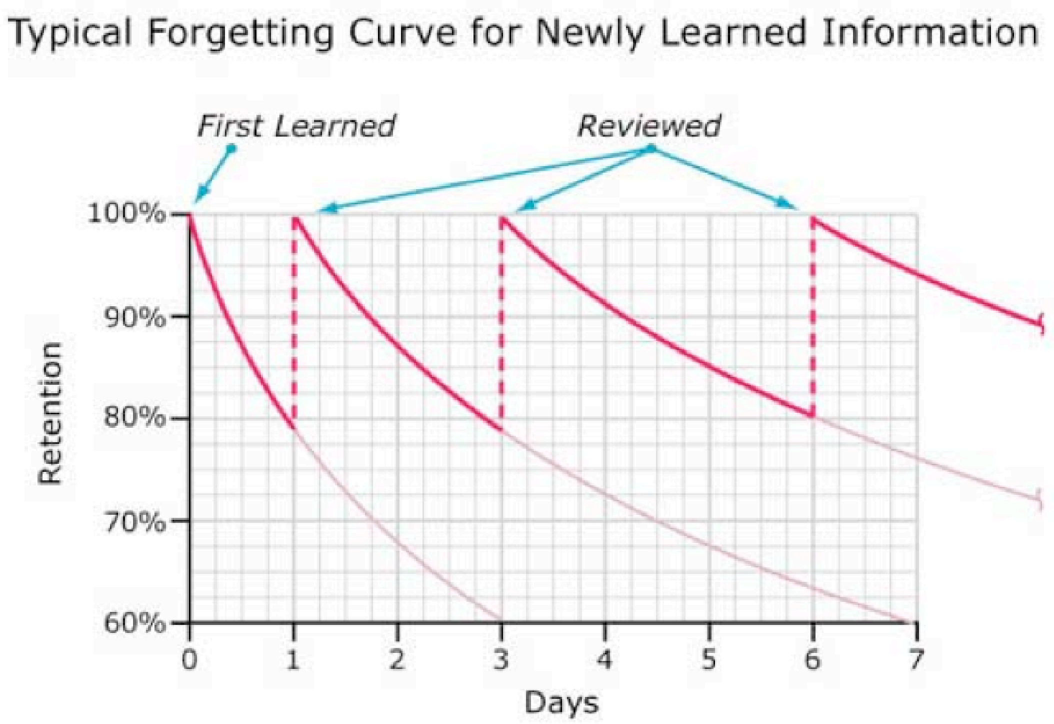
- memory weakens over time
- effortful retrieval strengthens access
- spacing makes retrieval effortful again

Interpretation

Spacing works not despite difficulty but because of it ([?Ebbinghaus1885]; [?BjorkBjork2011]).

Evidence





- Ebbinghaus established the basic empirical pattern: retention drops sharply without reinforcement ([?Ebbinghaus1885])
- distributed practice outperforms massed practice across many recall tasks ([?Cepeda2006])
- the practical point is *not* "wait as long as possible" but "revisit after enough delay that retrieval requires effort"

Repeat after some forgetting, not only before forgetting starts.

Doing Beats Hearing

Guiding Question

Why do interaction and explanation usually outperform listening alone?

$$\text{interactive} > \text{constructive} > \text{active} > \text{passive}$$

Why Might This Be True?

- passive listening creates little transformation
- constructive activity creates inferences
- interaction exposes misunderstandings

Interpretation

Discussion matters because it forces students to externalize, compare, revise, and defend understanding ([?Chi2009]).

Evidence

- **Passive:** hearing or reading without transformation
- **Active:** marking or selecting
- **Constructive:** generating explanations or examples
- **Interactive:** reasoning with another person and adapting to their response

> Durable learning requires doing, explaining, and interacting, not only hearing.

Studying With AI

Guiding Question

How should students use AI without weakening their own learning?

$$\text{tool support} + \text{retrieval} + \text{self-explanation} + \text{discussion}$$

Why Might This Be True?

- AI improves access to examples and reformulations
- AI can also create fluency illusions
- only retrieval and explanation reveal whether understanding is real

Interpretation

Use AI to generate contrasts, examples, and counterpositions. Then explain the concept *without* the tool and test whether the idea still holds together.

Evidence

- ask AI for alternative explanations, not final authority
- close the tool and reconstruct the idea from memory
- compare your answer with the AI output and identify what you missed
- discuss the mismatch with peers or in class

> Use AI as a sparring partner, not as a replacement for interpretation and retrieval.

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